

ASVH-SPEC-001

Runtime Authority Specification

Deterministic Runtime Evaluation for AI-enabled Ambient Scribing

Version: 1.0

Status: Engineering Specification

Project: ASVH-001 – NHS Ambient Scribing Runtime Verification Harness

1. Purpose

1.1 Overview

This specification defines the normative behaviour of the Runtime Authority implemented by the ASVH Runtime Verification Framework.

The Runtime Authority is an implementation-independent engineering component that evaluates whether delegated authority remains legitimately exercisable immediately before an AI-generated clinical artefact produces an institutional consequence.

Within the ASVH Reference Implementation, the institutional consequence is the commitment of AI-generated clinical documentation to an Electronic Patient Record (EPR).

This specification establishes the contractual behaviour required of a conforming Runtime Authority implementation. It defines the required runtime evaluation semantics independently of software architecture, programming language, deployment model or execution platform.

1.2 Specification Objectives

This specification defines:

- the Runtime Authority and its responsibilities;
- the Runtime Context consumed during evaluation;
- the Runtime Rule catalogue;
- deterministic Runtime Rule behaviour;
- Runtime Authority decision formation;
- Authority Receipt generation;
- conformance requirements for compliant implementations.

The specification intentionally does not define software architecture, user interface behaviour, persistence mechanisms or deployment technologies.

1.3 Normative Language

The key words **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT** and **MAY** are to be interpreted as normative requirements within this specification.

Where these terms are used, they describe the mandatory or optional behaviour expected of any Runtime Authority implementation claiming conformance with ASVH-SPEC-001.

1.4 Relationship to Companion Publications

ASVH-SPEC-001 forms the normative specification within the ASVH engineering publication series.

Publication	Engineering Responsibility
ASVH-METH-001	Defines the Runtime Requirement Engineering methodology.
ASVH-TRACE-001	Defines the engineering relationships between Runtime Verification artefacts.
ASVH-SPEC-001	Defines the normative behaviour of the Runtime Authority.
ASVH-ARCH-001	Defines the architectural placement of the Runtime Authority.
ASVH-IMPL-001	Demonstrates a conforming software implementation.

This specification assumes that Runtime Requirements and Runtime Context have already been derived using the engineering methodology described in **ASVH-METH-001** and that their engineering relationships are documented in **ASVH-TRACE-001**. Accordingly, this publication does not redefine those artefacts. Instead, it specifies the behaviour that the Runtime Authority **SHALL** exhibit when evaluating Runtime Context at the Execution Bind Point.

1.5 Engineering Scope

This specification defines the normative behaviour of the Runtime Authority.

It specifies:

- what information SHALL be evaluated;
- how Runtime Rules SHALL behave;
- how Runtime Authority decisions SHALL be formed;
- what evidence SHALL be produced;
- what constitutes a conforming Runtime Authority implementation.

It does **not** define:

- Runtime Requirement Engineering;
- engineering traceability methodology;
- software architecture;
- implementation technology;
- deployment models.

Those responsibilities are defined by the companion publications.

1.6 Engineering Summary

ASVH-SPEC-001 establishes the Runtime Authority as a deterministic, implementation-independent engineering component responsible for evaluating runtime admissibility immediately before institutional consequence formation.

By defining mandatory Runtime Rule behaviour, deterministic decision formation, Authority Receipt generation and conformance requirements, this specification provides the normative contract against which Runtime Authority implementations can be objectively assessed.

2. Scope

2.1 Specification Scope

This specification defines the normative behaviour of a Runtime Authority operating at the Execution Bind Point immediately before institutional consequence formation.

The specification applies only to deterministic runtime evaluation and the generation of engineering evidence supporting Runtime Authority decisions.

It establishes the minimum behavioural requirements that every conforming Runtime Authority implementation **SHALL** satisfy.

2.2 In Scope

A conforming Runtime Authority implementation **SHALL**:

- consume a valid Runtime Context;
 - evaluate the Runtime Context using the Runtime Rule Catalogue defined within this specification;
 - evaluate Runtime Rules deterministically;
 - produce exactly one Runtime Authority decision;
 - generate an Authority Receipt describing the completed evaluation;
 - preserve evidence supporting deterministic reconstruction of the evaluation;
 - operate independently of implementation technology.
-

2.3 Out of Scope

The Runtime Authority **SHALL NOT**:

- establish governance policy;
- determine organisational delegation;
- approve AI products for deployment;
- perform clinical assessment or diagnosis;
- determine treatment recommendations;
- replace professional clinical judgement;
- interpret legislation or regulation;
- replace organisational accountability;
- perform software orchestration beyond Runtime Authority evaluation.

These responsibilities remain outside the scope of the Runtime Authority and are expected to be fulfilled by organisational governance, clinical practice, regulatory frameworks or implementation-specific components.

2.4 Execution Boundary

The Runtime Authority operates exclusively at the **Execution Bind Point**.

Its responsibility begins when a valid Execution Request is presented for evaluation and ends when one of the following deterministic outcomes has been produced:

- **ALLOW**
- **ESCALATE**
- **REFUSE**

The Runtime Authority has no responsibility for events occurring before Runtime Context is presented or after the resulting Runtime Authority decision has been returned.

2.5 Architectural Separation

The Runtime Authority is an independent engineering component.

It **SHALL** evaluate only the Runtime Context presented at the point of execution.

It **SHALL NOT** modify the Runtime Context, alter organisational governance, or participate in downstream execution.

This separation preserves deterministic evaluation and ensures that Runtime Authority behaviour remains independent of implementation technology and operational workflow.

2.6 Relationship to Governance

The Runtime Authority complements organisational governance but does not replace it.

Governance establishes the organisational conditions under which authority may be delegated.

The Runtime Authority evaluates whether that delegated authority remains legitimately exercisable immediately before institutional consequence formation.

Accordingly, governance and Runtime Authority fulfil different engineering responsibilities and operate at different stages of the execution lifecycle.

2.7 Engineering Summary

The scope of this specification is intentionally narrow.

It defines only the deterministic behaviour required of a Runtime Authority operating at the Execution Bind Point.

All responsibilities relating to governance, clinical decision-making, organisational policy, software implementation and deployment remain outside the scope of this specification.

3. Runtime Authority

3.1 Definition

The Runtime Authority is an implementation-independent engineering component responsible for determining whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

It operates at the Execution Bind Point, where a completed execution request is evaluated against the Runtime Context presented at that moment.

The Runtime Authority produces a single deterministic Runtime Authority Decision based solely upon the Runtime Context evaluated during that execution.

3.2 Engineering Responsibility

A conforming Runtime Authority **SHALL**:

- consume the complete Runtime Context presented at the Execution Bind Point;
- evaluate the applicable Runtime Rules defined by this specification;
- determine a single Runtime Authority Decision;
- generate an Authority Receipt describing the completed evaluation;
- preserve deterministic behaviour throughout the evaluation process.

A Runtime Authority **SHALL NOT** perform responsibilities assigned to governance, implementation infrastructure or clinical decision-making.

3.3 Execution Bind Point

The Runtime Authority operates exclusively at the **Execution Bind Point**.

The Execution Bind Point represents the final deterministic evaluation boundary immediately before an execution request produces an institutional consequence.

No execution beyond the Execution Bind Point shall occur until the Runtime Authority has completed evaluation and returned one of the defined Runtime Authority Decisions.

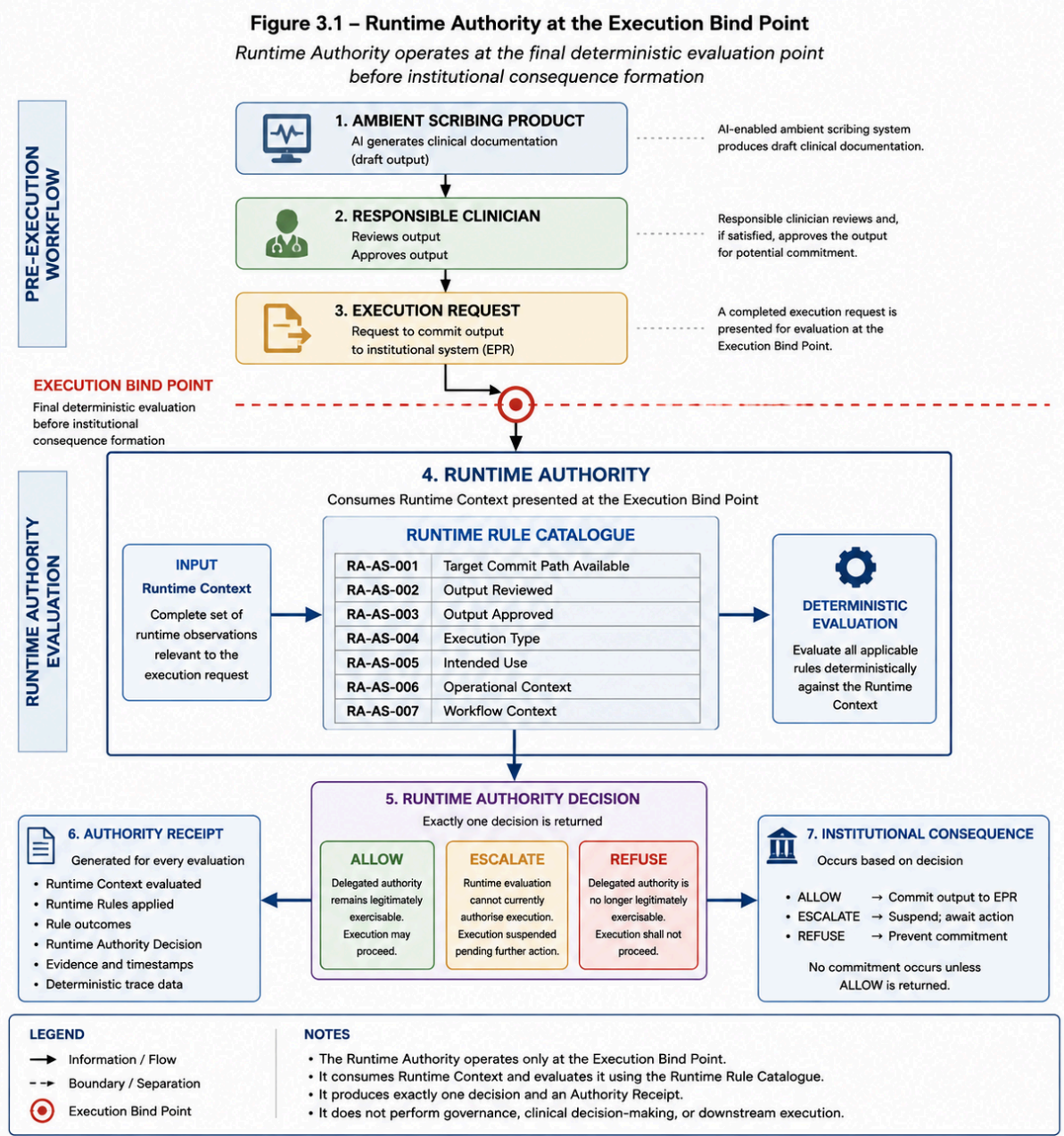


Figure 3.1 – Runtime Authority at the Execution Bind Point

Figure 3.1 defines the normative operating boundary of the Runtime Authority within the ASVH Runtime Verification Framework. The diagram illustrates the complete execution pathway from AI-assisted clinical documentation generation through to institutional consequence formation, identifying the **Execution Bind Point** as the sole location at which Runtime Authority evaluation occurs.

The upper portion of the figure represents the **pre-execution workflow**. An AI-enabled ambient scribing product generates draft clinical documentation, which is subsequently reviewed and approved by the responsible clinician in accordance with established clinical practice. Once clinician review is complete, an **Execution Request** is created requesting commitment of the approved documentation to the Electronic Patient Record (EPR). At this

stage, no institutional consequence has yet occurred, and execution remains suspended pending Runtime Authority evaluation.

The **Execution Bind Point** represents the final deterministic evaluation boundary before institutional consequence formation. It separates workflow preparation from consequence execution and defines the exclusive operating boundary of the Runtime Authority. No downstream execution is permitted to proceed until Runtime Authority evaluation has completed and a valid Runtime Authority Decision has been returned.

Within the Runtime Authority, the complete **Runtime Context** presented by the execution request is consumed as the input to deterministic evaluation. The Runtime Authority does not create, modify or infer Runtime Context; it evaluates only the observable operational state presented at the Execution Bind Point. The Runtime Context is assessed against the complete **Runtime Rule Catalogue**, comprising the seven Runtime Rules defined by this specification. Each Runtime Rule evaluates an independent aspect of the execution request, including commit path availability, clinician review, clinician approval, execution type, intended use, operational context and workflow context.

Following completion of Runtime Rule evaluation, the Runtime Authority performs a **deterministic evaluation** and returns exactly one permitted Runtime Authority Decision: **ALLOW**, **ESCALATE**, or **REFUSE**. These decision values are normative and represent the complete set of outcomes defined by this specification. No additional decision states are permitted.

Every completed evaluation results in the generation of an **Authority Receipt**, providing the authoritative engineering record of the Runtime Context evaluated, the Runtime Rules applied, the individual rule outcomes, the Runtime Authority Decision and the supporting evidence required for deterministic reconstruction of the evaluation. The Authority Receipt is generated irrespective of the decision returned and provides the implementation-independent evidence supporting Runtime Authority behaviour.

The final stage illustrates the relationship between the Runtime Authority Decision and the resulting institutional consequence. Only an **ALLOW** decision permits commitment of the clinical documentation to the Electronic Patient Record. An **ESCALATE** decision suspends execution pending further operational action or review, while a **REFUSE** decision prevents institutional consequence formation entirely. In all cases, the Runtime Authority determines only whether delegated authority remains legitimately exercisable immediately before execution; it does not perform governance, clinical decision-making or downstream execution.

The figure therefore establishes the Runtime Authority as a deterministic, implementation-independent evaluation component operating exclusively at the Execution Bind Point. By consuming Runtime Context, evaluating the Runtime Rule Catalogue, producing a single normative decision and generating an Authority Receipt, the Runtime Authority provides the engineering mechanism through which runtime admissibility is determined immediately before institutional consequence formation.

3.4 Runtime Evaluation Boundary

The Runtime Authority evaluates only the Runtime Context presented for the current execution request.

A conforming implementation:

- **SHALL** evaluate observable Runtime Context;
- **SHALL NOT** infer unavailable information;
- **SHALL NOT** modify Runtime Context during evaluation;
- **SHALL NOT** depend upon implementation-specific behaviour not represented within the Runtime Context.

This ensures that Runtime Authority decisions remain deterministic and reproducible.

3.5 Independence

The Runtime Authority is architecturally independent of:

- governance systems;
- ambient scribing products;
- Electronic Patient Record systems;
- workflow orchestration;
- downstream execution.

Architectural independence ensures that Runtime Authority decisions are derived exclusively from the Runtime Context presented during evaluation rather than from implementation-specific behaviours or external system state.

Independence within this specification refers to **architectural responsibility**, not organisational ownership or deployment topology.

3.6 Runtime Authority Decision

A conforming Runtime Authority **SHALL** produce exactly one Runtime Authority Decision for every valid evaluation request.

The permitted decisions are:

Decision	Behaviour
ALLOW	Delegated authority remains legitimately exercisable. Execution may proceed.
ESCALATE	Runtime evaluation cannot currently authorise execution. Execution is suspended pending further action.
REFUSE	Delegated authority is no longer legitimately exercisable. Execution shall not proceed.

No additional Runtime Authority Decision values are permitted by this specification.

3.7 Authority Receipt

Completion of every Runtime Authority evaluation **SHALL** result in the generation of an Authority Receipt.

The Authority Receipt constitutes the authoritative engineering record describing:

- the Runtime Context evaluated;
- the Runtime Rules applied;
- the Runtime Authority Decision;
- the evidence generated during evaluation.

The Authority Receipt requirements are specified in Chapter 10.

3.8 Deterministic Behaviour

A conforming Runtime Authority **SHALL** satisfy the following behavioural properties:

- identical Runtime Context shall produce identical Runtime Authority Decisions;
- Runtime Rule evaluation shall be deterministic;
- Runtime Context shall remain immutable during evaluation;
- Authority Receipt generation shall accurately represent the completed evaluation.

These properties define the minimum deterministic behaviour required by this specification.

3.9 Relationship to Companion Publications

This chapter defines the Runtime Authority itself.

Companion publications provide supporting engineering information.

Publication	Relationship
ASVH-METH-001	Defines how Runtime Requirements are engineered.
ASVH-TRACE-001	Defines the engineering relationships supporting Runtime Authority behaviour.
ASVH-ARCH-001	Describes the architectural placement of the Runtime Authority.
ASVH-IMPL-001	Demonstrates a conforming Runtime Authority implementation.

3.10 Engineering Summary

The Runtime Authority is the deterministic execution evaluation component defined by this specification.

Its sole responsibility is to determine whether delegated authority remains legitimately exercisable immediately before institutional consequence formation by evaluating Runtime Context against the Runtime Rule Catalogue.

By maintaining architectural independence, deterministic behaviour and implementation independence, the Runtime Authority provides a consistent and reproducible engineering mechanism for runtime admissibility evaluation.

4. Runtime Context Specification

4.1 Overview

The Runtime Authority SHALL evaluate a Runtime Context for every execution request presented at the Execution Bind Point.

Runtime Context represents the complete set of observable operational information required to determine whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

A Runtime Context SHALL be presented as a structured engineering artefact and SHALL contain all information required for deterministic Runtime Rule evaluation.

The Runtime Authority SHALL evaluate only the Runtime Context presented for the current execution request.

4.2 Engineering Purpose

Runtime Context provides the normative input to Runtime Authority evaluation.

A conforming Runtime Context SHALL:

- represent the operational state immediately before execution;
- contain only observable runtime information;
- support deterministic Runtime Rule evaluation;
- remain implementation independent;
- preserve evidence supporting Authority Receipt generation.

The Runtime Context SHALL NOT contain information that cannot be objectively evaluated during runtime.

4.3 Runtime Context Characteristics

A conforming Runtime Context SHALL exhibit the following characteristics.

Characteristic	Requirement
Complete	SHALL contain every Runtime Context element required by the applicable Runtime Rules.
Observable	SHALL contain only operational state available at the Execution Bind Point.
Immutable	SHALL NOT be modified during Runtime Authority evaluation.
Deterministic	SHALL produce identical Runtime Rule outcomes when identical Runtime Context is presented.
Implementation Independent	SHALL preserve identical engineering semantics regardless of implementation technology.

4.4 Runtime Context Domains

The Runtime Context defined by this specification is composed of five engineering domains.

Domain	Purpose
Consultation Context	Represents the clinical consultation associated with the execution request.
Output Context	Represents the approval status of the AI-generated clinical documentation.

Execution Context Represents the requested institutional consequence.

Operational Context Represents operational conditions influencing execution.

Workflow Context Represents the workflow state immediately before execution.

Each Runtime Context domain contributes one or more Runtime Context elements used by the Runtime Rule Catalogue.

4.5 Runtime Context Requirements

Every Runtime Context element SHALL satisfy the following requirements.

A Runtime Context element:

- SHALL possess a unique Runtime Context Identifier;
- SHALL have a defined engineering meaning;
- SHALL identify its source;
- SHALL specify its data type;
- SHALL identify the Runtime Rule(s) consuming it;
- SHALL identify the corresponding Authority Receipt mapping.

Runtime Context elements that do not satisfy these requirements SHALL NOT be considered conformant with this specification.

4.6 Runtime Context Lifecycle

The Runtime Context lifecycle defined by this specification consists of five stages.

1. Runtime Context assembled.
2. Runtime Context validated.
3. Runtime Context presented to the Runtime Authority.
4. Runtime Context evaluated.
5. Runtime Context preserved within the Authority Receipt.

Once presented to the Runtime Authority, the Runtime Context SHALL remain immutable until evaluation has completed.

4.7 Relationship to Runtime Rules

The Runtime Context provides the sole engineering input to Runtime Rule evaluation.

Runtime Rules SHALL evaluate only Runtime Context elements defined by this specification.

Runtime Rules SHALL NOT obtain evaluation data from implementation-specific mechanisms that are not represented within the Runtime Context.

This requirement preserves deterministic behaviour and implementation independence.

4.8 Relationship to Authority Receipt

Following Runtime Authority evaluation, the Runtime Context SHALL be represented within the generated Authority Receipt.

The Authority Receipt SHALL preserve sufficient Runtime Context information to enable deterministic reconstruction of the completed evaluation.

Authority Receipt requirements are specified in Chapter 10.

4.9 Relationship to Companion Publications

This chapter defines the normative Runtime Context consumed by the Runtime Authority.

Companion publications provide additional engineering information.

Publication	Relationship
ASVH-METH-001	Defines the methodology used to derive Runtime Context.
ASVH-TRACE-001	Defines the engineering relationships between Runtime Context and the remaining engineering artefacts.
ASVH-ARCH-001	Describes the architectural placement of Runtime Context within the Runtime Verification Framework.

4.10 Engineering Summary

Runtime Context constitutes the normative input consumed by the Runtime Authority.

By defining a complete, observable, immutable and implementation-independent representation of the operational state immediately before institutional consequence formation, Runtime Context provides the deterministic engineering foundation upon which all Runtime Rule evaluation defined by this specification depends.

5. Runtime Context Catalogue

5.1 Overview

The Runtime Context Catalogue defines the normative Runtime Context elements that may be presented to the Runtime Authority during evaluation.

Each Runtime Context element possesses:

- a unique Runtime Context Identifier;
- a defined engineering purpose;
- a specified data type;
- a permitted value domain;
- one or more consuming Runtime Rules;
- an Authority Receipt mapping.

Only Runtime Context elements defined within this catalogue SHALL be considered conformant with this specification.

5.2 Runtime Context Catalogue Structure

Each Runtime Context element SHALL be specified using the following structure.

Property	Description
Runtime Context Identifier	Unique identifier
Name	Engineering name
Purpose	Engineering meaning
Data Type	Boolean, String, Enumeration, etc.
Required	Yes / No

Source	Originating system or process
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Consumed By	Runtime Rule(s)
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Authority Receipt Mapping	Receipt field
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Deterministic Behaviour	Required evaluation characteristics
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5.3 Consultation Context

RC-001 Consultation Active

Property	Value
Identifier	RC-001
Name	Consultation Active
Purpose	Indicates that the consultation associated with the execution request remains active.
Data Type	Boolean
Required	Yes

Source	Clinical Workflow
Consumed By	RA-AS-007
Authority Receipt	Consultation Status
Deterministic Behaviour	SHALL evaluate identically for identical Runtime Context.

RC-002 Clinician Authenticated

Property	Value
Identifier	RC-002
Name	Clinician Authenticated
Purpose	Confirms that the responsible clinician has been authenticated for the current consultation.
Data Type	Boolean
Required	Yes
Source	Identity Service
Consumed By	RA-AS-007

Authority Receipt	Authentication Status
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Deterministic Behaviour	SHALL evaluate identically for identical Runtime Context.
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5.4 Output Context

RC-003 Output Reviewed

Property	Value
Identifier	RC-003
Name	Output Reviewed
Purpose	Indicates clinician review has completed.
Data Type	Boolean
Required	Yes
Source	Clinical Workflow
Consumed By	RA-AS-002

RC-004 Output Approved

Property	Value
Identifier	RC-004
Name	Output Approved
Purpose	Indicates clinician approval has been completed.
Data Type	Boolean
Required	Yes
Source	Clinical Workflow
Consumed By	RA-AS-003
Authority Receipt	Approval Status

5.5 Operational Context

RC-005 Intended Use

Property	Value
Identifier	RC-005
Name	Intended Use
Purpose	Confirms execution remains within the approved operational scope.
Data Type	Enumeration
Required	Yes
Source	Operational Policy
Consumed By	RA-AS-005
Authority Receipt	Intended Use Status

RC-006 Commit Path Available

Property	Value
Identifier	RC-006

Name	Commit Path Available
Purpose	Confirms that the requested execution target remains available.
Data Type	Boolean
Required	Yes
Source	Target System
Consumed By	RA-AS-001
Authority Receipt	Commit Path Status

5.6 Workflow Context

RC-007 Workflow Context

Property	Value
Identifier	RC-007
Name	Workflow Context

Purpose	Represents the workflow state immediately before execution.
Data Type	Enumeration
Required	Yes
Source	Workflow Engine
Consumed By	RA-AS-007
Authority Receipt	Workflow Status

5.7 Execution Context

RC-008 Execution Type

Property	Value
Identifier	RC-008
Name	Execution Type
Purpose	Identifies the requested institutional consequence.

Data Type	Enumeration
Required	Yes
Source	Execution Request
Consumed By	RA-AS-004
Authority Receipt	Execution Type

RC-009 Operational Context

Property	Value
Identifier	RC-009
Name	Operational Context
Purpose	Represents runtime operational conditions affecting execution.
Data Type	Enumeration
Required	Yes
Source	Runtime Environment

Consumed By RA-AS-006

Authority Receipt Operational Context

5.8 Catalogue Integrity

Every Runtime Context element defined within this catalogue SHALL:

- possess a unique identifier;
- have one engineering meaning;
- be consumed by one or more Runtime Rules;
- map to Authority Receipt evidence;
- remain immutable during Runtime Authority evaluation.

No additional Runtime Context elements SHALL be considered conformant unless incorporated into a future revision of this specification.

5.9 Engineering Summary

The Runtime Context Catalogue defines the complete set of Runtime Context elements recognised by this specification.

These elements constitute the exclusive engineering input to Runtime Rule evaluation and provide the deterministic operational state consumed by the Runtime Authority. Every Runtime Context element is uniquely identified, normatively specified and directly traceable to the Runtime Rule Catalogue and Authority Receipt defined in subsequent chapters.

6. Runtime Rule Catalogue

6.1 Overview

The Runtime Rule Catalogue defines the complete set of Runtime Rules recognised by this specification.

Each Runtime Rule evaluates one specific aspect of the Runtime Context presented at the Execution Bind Point.

Collectively, the Runtime Rules determine whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

A conforming Runtime Authority **SHALL** evaluate only Runtime Rules defined within this catalogue.

No additional Runtime Rules SHALL be considered conformant unless introduced by a future revision of this specification.

6.2 Engineering Purpose

The Runtime Rule Catalogue provides the normative behavioural definition of Runtime Authority evaluation.

Each Runtime Rule:

- evaluates one engineering condition;
- consumes one or more Runtime Context elements;
- produces one deterministic Runtime Rule outcome;
- contributes to exactly one Runtime Authority Decision;
- records its evaluation within the Authority Receipt.

Runtime Rules SHALL be evaluated independently.

6.3 Runtime Rule Characteristics

Every Runtime Rule defined by this specification SHALL exhibit the following characteristics.

Characteristic	Requirement
Deterministic	Identical Runtime Context SHALL produce identical Runtime Rule outcomes.

Independent	Runtime Rules SHALL evaluate independently of one another.
Observable	Runtime Rules SHALL evaluate only Runtime Context defined by this specification.
Implementation Independent	Runtime Rule behaviour SHALL remain identical across implementations.
Traceable	Every Runtime Rule SHALL map to Authority Receipt evidence.

6.4 Runtime Rule Catalogue

The Runtime Rule Catalogue defined by this specification comprises seven Runtime Rules.

Rule Identifier	Runtime Rule	Runtime Context	Primary Purpose	Outcome
RA-AS-001	Target Commit Path Available	RC-006	Confirm the requested commit path remains available.	ALLOW / ESCALATE
RA-AS-002	Output Reviewed	RC-003	Confirm clinician review has completed.	ALLOW / REFUSE
RA-AS-003	Output Approved	RC-004	Confirm clinician approval has been completed.	ALLOW / REFUSE
RA-AS-004	Execution Type Identified	RC-008	Confirm the requested execution type is recognised.	ALLOW / REFUSE
RA-AS-005	Intended Use Maintained	RC-005	Confirm execution remains within approved intended use.	ALLOW / REFUSE
RA-AS-006	Operational Context Valid	RC-009	Confirm operational conditions remain valid.	ALLOW / ESCALATE

RA-AS-007	Workflow Context Valid	RC-001, RC-002, RC-007	Confirm the workflow state supports execution.	ALLOW / REFUSE
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Each Runtime Rule is specified normatively in Chapter 7.

6.5 Runtime Rule Evaluation Principles

A conforming Runtime Authority SHALL evaluate Runtime Rules according to the following principles.

- Every applicable Runtime Rule SHALL be evaluated.
- Runtime Rules SHALL evaluate only Runtime Context presented for the current execution request.
- Runtime Rules SHALL NOT modify Runtime Context.
- Runtime Rules SHALL NOT depend upon implementation-specific behaviour outside the Runtime Context.
- Runtime Rules SHALL produce deterministic outcomes.

These principles establish the minimum behavioural requirements for Runtime Rule evaluation.

6.6 Relationship to Runtime Context

Runtime Rules consume Runtime Context.

Every Runtime Rule SHALL identify:

- the Runtime Context required;
- the Runtime Context identifiers consumed;
- the Authority Receipt fields generated.

Runtime Rules SHALL NOT evaluate Runtime Context that is not defined by this specification.

6.7 Relationship to Runtime Authority Decisions

Runtime Rule outcomes contribute collectively to the Runtime Authority Decision.

A Runtime Rule does not independently authorise or prevent execution.

Instead, each Runtime Rule contributes objective engineering evidence that is considered by the Runtime Authority during deterministic decision formation.

The Runtime Authority Decision Model is defined in Chapter 9.

6.8 Relationship to Authority Receipt

Completion of every Runtime Rule evaluation SHALL be recorded within the Authority Receipt.

The Authority Receipt SHALL preserve sufficient information to reconstruct:

- the Runtime Rule evaluated;
- the Runtime Context consumed;
- the Runtime Rule outcome;
- the associated engineering evidence.

Authority Receipt requirements are specified in Chapter 10.

6.9 Relationship to Companion Publications

This chapter defines the Runtime Rule Catalogue recognised by the Runtime Authority.

Supporting engineering information is provided by the companion publications.

Publication	Relationship
ASVH-METH-001	Defines the methodology used to derive Runtime Rules.
ASVH-TRACE-001	Defines the engineering relationships linking Runtime Rules to Runtime Context, Authority Receipts and Verification Scenarios.
ASVH-ARCH-001	Describes the architectural placement of Runtime Rule evaluation.
ASVH-IMPL-001	Demonstrates one conforming implementation of the Runtime Rule Catalogue.

6.10 Engineering Summary

The Runtime Rule Catalogue defines the complete set of deterministic Runtime Rules recognised by this specification.

Each Runtime Rule evaluates a single engineering condition using the Runtime Context presented at the Execution Bind Point. Collectively, the Runtime Rule Catalogue establishes the normative behavioural contract upon which Runtime Authority evaluation is based.

The detailed normative specification for each Runtime Rule is provided in **Chapter 7**.

7. Runtime Rule Specifications

7.1 Overview

This chapter defines the normative behaviour of every Runtime Rule recognised by this specification.

Runtime Rules represent the deterministic engineering logic through which the Runtime Authority evaluates the Runtime Context presented at the Execution Bind Point. Each Runtime Rule evaluates a specific observable runtime condition and produces an objective engineering outcome that contributes to the overall Runtime Authority Decision.

A conforming Runtime Authority **SHALL** evaluate Runtime Rules only in accordance with the normative definitions contained within this chapter.

Every Runtime Rule defined by this specification:

- SHALL evaluate only the Runtime Context identified within the rule specification;
- SHALL exhibit deterministic behaviour;
- SHALL produce only the outcomes defined by this specification;
- SHALL contribute engineering evidence to the Authority Receipt;
- SHALL preserve implementation-independent behaviour.

No implementation SHALL alter the engineering meaning of a Runtime Rule defined within this specification.

Figure 7.1 – Runtime Rule Evaluation Structure

The Runtime Authority evaluates each Runtime Rule using a consistent deterministic structure, generating evidence that maps to the Authority Receipt.

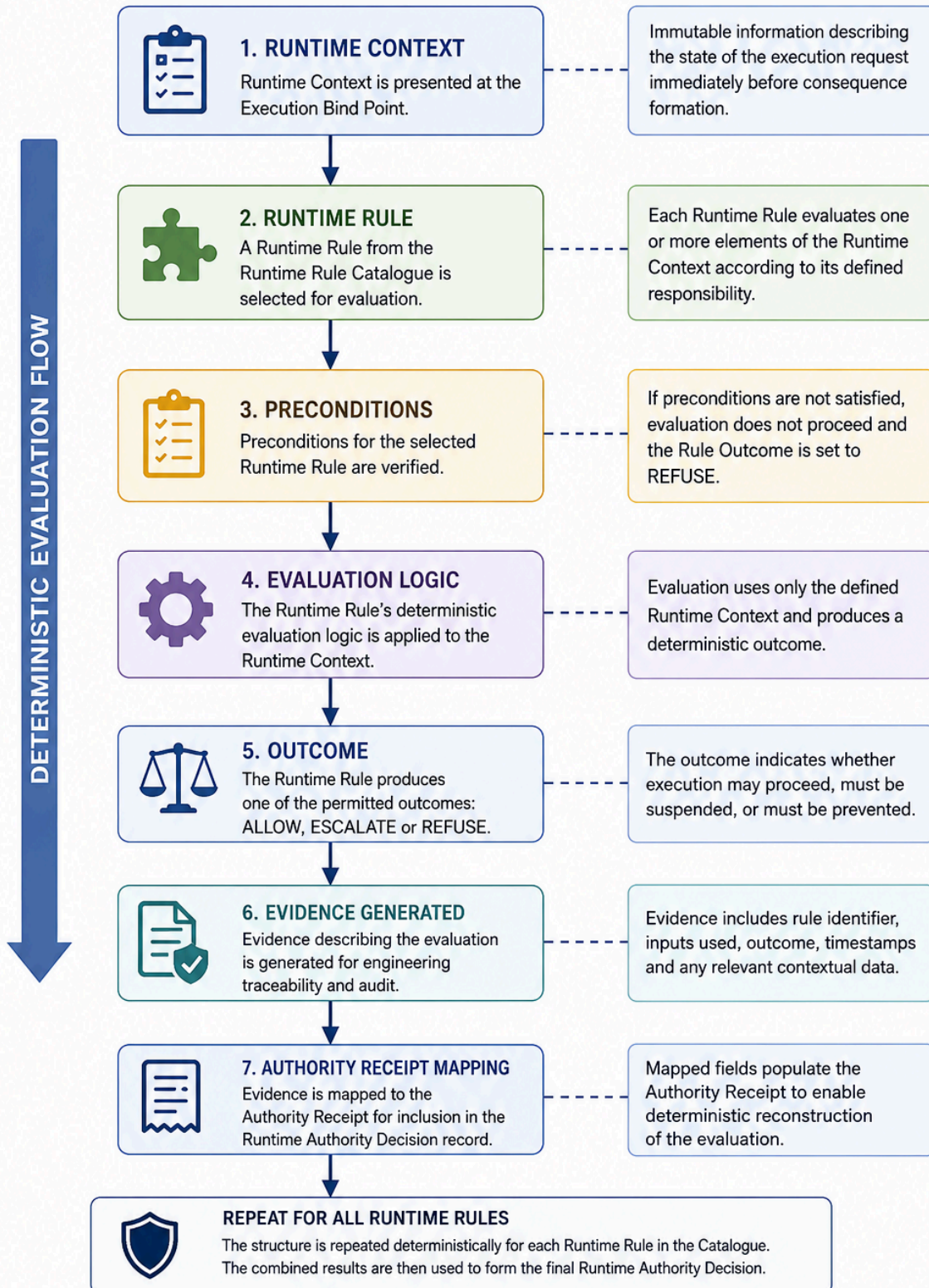


Figure 7.1 – Runtime Rule Evaluation Structure

Figure 7.1 illustrates the consistent normative structure applied to every Runtime Rule defined within ASVH-SPEC-001. The diagram shows how each rule begins with the immutable Runtime Context presented at the Execution Bind Point, then proceeds through rule selection, precondition validation, deterministic evaluation logic, outcome formation, evidence generation and Authority Receipt mapping.

The first stage establishes the **Runtime Context** as the sole input to the Runtime Rule. This context represents the observable operational state immediately before consequence formation and remains unchanged throughout evaluation. The Runtime Authority then selects the applicable **Runtime Rule** from the Runtime Rule Catalogue according to the rule's defined engineering responsibility.

Before evaluation proceeds, the rule's **Preconditions** are checked. These conditions confirm that the execution request, required Runtime Context and other mandatory inputs are valid and available. Where preconditions are not satisfied, evaluation does not continue in accordance with the rule's defined behaviour.

The Runtime Authority then applies the rule's **Evaluation Logic** deterministically to the relevant Runtime Context. The rule produces one permitted **Outcome**—ALLOW, ESCALATE or REFUSE—according to the normative logic defined in Chapter 7.

Each completed evaluation generates structured **Engineering Evidence**, including the Runtime Rule identifier, inputs evaluated, rule outcome, timestamps and relevant contextual information. That evidence is then mapped into the **Authority Receipt**, preserving the information required for engineering traceability, audit and deterministic reconstruction of the evaluation.

The complete structure is repeated consistently for every Runtime Rule in the catalogue. The combined Runtime Rule outcomes are subsequently used by the Runtime Evaluation Engine to form the final Runtime Authority Decision.

7.2 Runtime Rule Specification Structure

Each Runtime Rule defined within this specification is described using the same normative structure.

Section	Purpose
Identifier	Unique Runtime Rule Identifier
Name	Runtime Rule Name
Purpose	Engineering objective of the Runtime Rule
Runtime Context	Runtime Context consumed during evaluation
Preconditions	Conditions that SHALL exist before evaluation
Required Inputs	Runtime Context elements evaluated
Evaluation Logic	Deterministic evaluation behaviour
Permitted Outcomes	Runtime Rule outcomes permitted by this specification
Evidence Generated	Engineering evidence produced during evaluation
Authority Receipt Mapping	Information preserved within the Authority Receipt
Deterministic Behaviour	Required deterministic properties
Conformance Requirements	Mandatory implementation behaviour

All Runtime Rules SHALL conform to this structure.

7.3 RA-AS-001 — Target Commit Path Available

Identifier

RA-AS-001

Name

Target Commit Path Available

Purpose

The purpose of Runtime Rule **RA-AS-001** is to determine whether the requested institutional destination remains capable of accepting the execution request immediately before institutional consequence formation.

The Runtime Authority SHALL verify that a valid execution path exists before execution is permitted to proceed.

Runtime Context

This Runtime Rule consumes the following Runtime Context.

Runtime Context	Identifier
Commit Path Available	RC-006

Preconditions

Before Runtime Rule evaluation commences, the following conditions SHALL exist.

- A valid Execution Request SHALL have been received.
- Runtime Context SHALL have been successfully validated.
- Runtime Authority evaluation SHALL have commenced.
- Evaluation SHALL occur at the Execution Bind Point.

If these conditions are not satisfied, Runtime Rule evaluation SHALL NOT proceed.

Required Inputs

Runtime Context	Data Type	Required
Commit Path Available	Boolean	Yes
Target System	String	Yes
Execution Type	Enumeration	Yes

Evaluation Logic

The Runtime Authority SHALL evaluate Commit Path Availability according to the following deterministic logic.

The Runtime Rule SHALL NOT return REFUSE.

IF CommitPathAvailable = TRUE

THEN

Rule Outcome = ALLOW

ELSE

Rule Outcome = ESCALATE

Where the Runtime Authority cannot determine that the requested execution destination remains available, execution SHALL remain suspended until a subsequent Runtime Authority evaluation is completed.

Permitted Outcomes

Outcome	Behaviour
ALLOW	A valid execution destination exists. Runtime Rule evaluation succeeds.
ESCALATE	Execution destination cannot currently be confirmed. Runtime Authority evaluation continues in accordance with the Runtime Evaluation Engine.

Evidence Generated

Successful evaluation of Runtime Rule RA-AS-001 SHALL produce the following engineering evidence.

- Runtime Rule Identifier
 - Runtime Rule Version
 - Evaluation Timestamp
 - Commit Path Status
 - Target System Identifier
 - Runtime Rule Outcome
-

Authority Receipt Mapping

The Authority Receipt SHALL contain the following information derived from Runtime Rule RA-AS-001.

Authority Receipt Field	Source
Runtime Rule Identifier	RA-AS-001
Commit Path Status	RC-006
Runtime Rule Outcome	Evaluation Result
Evaluation Timestamp	Runtime Evaluation Engine

Deterministic Behaviour

Runtime Rule RA-AS-001 SHALL satisfy the following deterministic properties.

- Identical Runtime Context SHALL produce identical Runtime Rule outcomes.
- Runtime Context SHALL NOT be modified during evaluation.
- Evaluation SHALL be independent of implementation technology.
- Evaluation SHALL be reproducible from the Authority Receipt.

Conformance Requirements

A Runtime Authority implementation claiming conformance with this specification:

- SHALL evaluate RA-AS-001 before institutional consequence formation.
- SHALL evaluate only the Runtime Context identified within this specification.
- SHALL preserve Runtime Context throughout evaluation.
- SHALL record the Runtime Rule outcome within the Authority Receipt.
- SHALL produce deterministic behaviour.
- SHALL NOT return outcomes other than those defined by this specification.

Runtime Rule Summary

Runtime Rule RA-AS-001 confirms that a valid execution destination remains available immediately before institutional consequence formation. By ensuring that execution proceeds only where a valid institutional commit path exists, the Runtime Rule contributes objective engineering evidence supporting deterministic Runtime Authority evaluation.

7.4 RA-AS-002 — Output Reviewed

Identifier

RA-AS-002

Name

Output Reviewed

Purpose

The purpose of Runtime Rule **RA-AS-002** is to determine whether the AI-generated clinical documentation has been reviewed by the responsible clinician before institutional consequence formation.

The Runtime Authority SHALL verify that human review has occurred prior to execution. Runtime Rule RA-AS-002 does not assess the clinical correctness of the documentation; it evaluates only whether the required review activity has been completed.

Runtime Context

This Runtime Rule consumes the following Runtime Context.

Runtime Context	Identifier
-----------------	------------

Output Reviewed	RC-003
-----------------	--------

Consultation Active	RC-001
---------------------	--------

Preconditions

Before Runtime Rule evaluation commences, the following conditions SHALL exist.

- A valid Execution Request SHALL have been received.
- Runtime Context SHALL have been successfully validated.
- The associated consultation SHALL remain active.
- Runtime Authority evaluation SHALL have commenced.

If these conditions are not satisfied, Runtime Rule evaluation SHALL NOT proceed.

Required Inputs

Runtime Context	Data Type	Required
-----------------	-----------	----------

Output Reviewed	Boolean	Yes
-----------------	---------	-----

Consultation Active	Boolean	Yes
---------------------	---------	-----

Review Timestamp	DateTime	No
------------------	----------	----

Reviewer Identifier	String	No
---------------------	--------	----

Evaluation Logic

The Runtime Authority SHALL evaluate clinician review according to the following deterministic logic.

IF OutputReviewed = TRUE

THEN

Rule Outcome = ALLOW

ELSE

Rule Outcome = REFUSE

Runtime Rule RA-AS-002 SHALL NOT return ESCALATE.

Where clinician review has not been completed, institutional consequence SHALL NOT be permitted.

Permitted Outcomes

Outcome	Behaviour
ALLOW	Required clinician review has been completed.
REFUSE	Required clinician review has not been completed.

Evidence Generated

Successful evaluation SHALL generate:

- Runtime Rule Identifier
- Runtime Rule Version
- Review Status

- Review Timestamp (where available)
 - Reviewer Identifier (where available)
 - Evaluation Timestamp
 - Runtime Rule Outcome
-

Authority Receipt Mapping

Authority Receipt Field	Source
Runtime Rule Identifier	RA-AS-002
Review Status	RC-003
Review Timestamp	Runtime Context
Runtime Rule Outcome	Evaluation Result
Evaluation Timestamp	Runtime Evaluation Engine

Deterministic Behaviour

Runtime Rule RA-AS-002 SHALL satisfy the following properties.

- Identical Runtime Context SHALL produce identical Runtime Rule outcomes.
 - Runtime Context SHALL remain immutable during evaluation.
 - Runtime Rule evaluation SHALL be reproducible.
 - Evaluation SHALL be implementation independent.
-

Conformance Requirements

A conforming Runtime Authority:

- SHALL evaluate clinician review before execution.
 - SHALL evaluate only Runtime Context defined by this specification.
 - SHALL record the Runtime Rule outcome within the Authority Receipt.
 - SHALL preserve deterministic behaviour.
 - SHALL NOT infer review completion from information outside the Runtime Context.
-

Runtime Rule Summary

Runtime Rule RA-AS-002 confirms that the required clinician review has been completed before execution proceeds. The Runtime Rule evaluates only the existence of review and does not assess the quality or clinical correctness of that review.

7.5 RA-AS-003 — Output Approved

Identifier

RA-AS-003

Name

Output Approved

Purpose

The purpose of Runtime Rule **RA-AS-003** is to determine whether the responsible clinician has explicitly approved the AI-generated clinical documentation immediately before institutional consequence formation.

Approval represents the formal authorisation for execution and SHALL be distinguished from review.

Runtime Context

Runtime Context	Identifier
Output Approved	RC-004
Clinician Authenticated	RC-002

Preconditions

Before Runtime Rule evaluation commences:

- Runtime Context SHALL have been validated.
- The responsible clinician SHALL remain authenticated.
- Runtime Authority evaluation SHALL have commenced.
- A valid Execution Request SHALL exist.

If these conditions are not satisfied, Runtime Rule evaluation SHALL NOT proceed.

Required Inputs

Runtime Context	Data Type	Required
Output Approved	Boolean	Yes
Clinician Authenticated	Boolean	Yes
Approval Timestamp	DateTime	No
Approver Identifier	String	No

Evaluation Logic

The Runtime Authority SHALL evaluate clinician approval according to the following deterministic logic.

IF OutputApproved = TRUE

THEN

Rule Outcome = ALLOW

ELSE

Rule Outcome = REFUSE

Runtime Rule RA-AS-003 SHALL NOT return ESCALATE.

Absence of explicit approval SHALL prevent institutional consequence formation.

Permitted Outcomes

Outcome	Behaviour
---------	-----------

ALLOW	Explicit clinician approval has been completed.
--------------	---

REFUSE	Explicit clinician approval has not been completed.
---------------	---

Evidence Generated

Successful evaluation SHALL generate:

- Runtime Rule Identifier
- Runtime Rule Version
- Approval Status
- Approval Timestamp (where available)
- Approver Identifier (where available)

- Evaluation Timestamp
 - Runtime Rule Outcome
-

Authority Receipt Mapping

Authority Receipt Field	Source
-------------------------	--------

Runtime Rule Identifier	RA-AS-003
-------------------------	-----------

Approval Status	RC-004
-----------------	--------

Approval Timestamp	Runtime Context
--------------------	-----------------

Runtime Rule Outcome	Evaluation Result
----------------------	-------------------

Evaluation Timestamp	Runtime Evaluation Engine
----------------------	---------------------------

Deterministic Behaviour

Runtime Rule RA-AS-003 SHALL satisfy the following properties.

- Identical Runtime Context SHALL produce identical Runtime Rule outcomes.
 - Runtime Context SHALL remain immutable.
 - Evaluation SHALL be deterministic.
 - Evaluation SHALL be implementation independent.
-

Conformance Requirements

A conforming Runtime Authority:

- SHALL verify explicit clinician approval before execution.
- SHALL preserve Runtime Context during evaluation.

- SHALL record approval status within the Authority Receipt.
 - SHALL NOT infer approval from review status.
 - SHALL produce deterministic Runtime Rule outcomes.
-

Runtime Rule Summary

Runtime Rule RA-AS-003 confirms that explicit clinician approval exists immediately before institutional consequence formation. Review alone is insufficient. Institutional execution SHALL only proceed where explicit approval has been successfully evaluated by the Runtime Authority.

7.6 RA-AS-004 — Execution Type Identified

Identifier

RA-AS-004

Name

Execution Type Identified

Purpose

The purpose of Runtime Rule **RA-AS-004** is to verify that the requested institutional consequence is explicitly identified and recognised by the Runtime Authority before execution proceeds.

Every execution request SHALL identify the institutional action being requested. The Runtime Authority SHALL evaluate only recognised execution types defined within the implementation's authorised execution model.

This Runtime Rule does not determine whether execution is permitted; it determines whether the requested execution can be unambiguously evaluated.

Runtime Context

Runtime Context	Identifier
Execution Type	RC-008

Preconditions

Before Runtime Rule evaluation commences:

- A valid Execution Request SHALL exist.
- Runtime Context SHALL have been validated.
- The requested execution SHALL identify an execution type.
- Runtime Authority evaluation SHALL have commenced.

If these conditions are not satisfied, Runtime Rule evaluation SHALL NOT proceed.

Required Inputs

Runtime Context	Data Type	Required
-----------------	-----------	----------

Execution Type	Enumeration	Yes
----------------	-------------	-----

Execution Request Identifier	String	Yes
------------------------------	--------	-----

Evaluation Logic

The Runtime Authority SHALL evaluate the requested execution according to the following deterministic logic.

IF $\text{ExecutionType} \in \text{SupportedExecutionTypes}$

THEN

Rule Outcome = ALLOW

ELSE

Rule Outcome = REFUSE

Execution requests whose execution type cannot be recognised SHALL NOT proceed.

Permitted Outcomes

Outcome	Behaviour
---------	-----------

ALLOW	Execution type is recognised by the Runtime Authority.
--------------	--

REFUSE	Execution type is unknown or unsupported.
---------------	---

Runtime Rule RA-AS-004 SHALL NOT return ESCALATE.

Evidence Generated

Successful evaluation SHALL generate:

- Runtime Rule Identifier
 - Runtime Rule Version
 - Execution Type
 - Execution Request Identifier
 - Evaluation Timestamp
 - Runtime Rule Outcome
-

Authority Receipt Mapping

Authority Receipt Field	Source
Runtime Rule Identifier	RA-AS-004
Execution Type	RC-008
Runtime Rule Outcome	Evaluation Result
Evaluation Timestamp	Runtime Evaluation Engine

Deterministic Behaviour

Runtime Rule RA-AS-004 SHALL satisfy the following properties.

- Identical Runtime Context SHALL produce identical outcomes.
- Evaluation SHALL be deterministic.
- Runtime Context SHALL remain immutable.

- Behaviour SHALL remain implementation independent.
-

Conformance Requirements

A conforming Runtime Authority:

- SHALL evaluate the execution type before execution.
 - SHALL evaluate only recognised execution types.
 - SHALL preserve Runtime Context.
 - SHALL record the Runtime Rule outcome within the Authority Receipt.
 - SHALL NOT infer execution type from implementation-specific behaviour.
-

Runtime Rule Summary

Runtime Rule RA-AS-004 ensures that the institutional consequence requested by the execution request is explicitly identified before Runtime Authority evaluation proceeds. Unknown or unsupported execution types SHALL prevent execution.

7.7 RA-AS-005 — Intended Use Maintained

Identifier

RA-AS-005

Name

Intended Use Maintained

Purpose

The purpose of Runtime Rule **RA-AS-005** is to determine whether the requested execution remains within the authorised intended use of the AI-enabled ambient scribing system at the moment of execution.

The Runtime Authority SHALL evaluate whether the execution request continues to fall within the operational scope approved for the system. This Runtime Rule evaluates operational admissibility rather than software capability.

Runtime Context

Runtime Context	Identifier
Intended Use	RC-005

Preconditions

- Before Runtime Rule evaluation commences:
- Runtime Context SHALL have been validated.
 - Intended Use SHALL be present.
 - Runtime Authority evaluation SHALL have commenced.
-

Required Inputs

Runtime Context	Data Type	Required
Intended Use	Enumeration	Yes
Approved Operational Scope	Enumeration	Yes

Evaluation Logic

The Runtime Authority SHALL evaluate intended use according to the following deterministic logic.

IF IntendedUse = ApprovedOperationalScope

THEN

Rule Outcome = ALLOW

ELSE

Rule Outcome = REFUSE

Execution outside the approved intended use SHALL NOT be permitted.

Permitted Outcomes

Outcome	Behaviour
---------	-----------

ALLOW	Execution remains within the approved intended use.
--------------	---

REFUSE	Execution falls outside the approved intended use.
---------------	--

Runtime Rule RA-AS-005 SHALL NOT return ESCALATE.

Evidence Generated

The Runtime Authority SHALL generate:

- Runtime Rule Identifier
- Runtime Rule Version

- Intended Use
 - Approved Operational Scope
 - Runtime Rule Outcome
 - Evaluation Timestamp
-

Authority Receipt Mapping

Authority Receipt Field	Source
-------------------------	--------

Runtime Rule Identifier	RA-AS-005
-------------------------	-----------

Intended Use	RC-005
--------------	--------

Runtime Rule Outcome	Evaluation Result
----------------------	-------------------

Evaluation Timestamp	Runtime Evaluation Engine
----------------------	---------------------------

Deterministic Behaviour

Runtime Rule RA-AS-005 SHALL satisfy the following properties.

- Evaluation SHALL be deterministic.
 - Runtime Context SHALL remain immutable.
 - Identical Runtime Context SHALL produce identical Runtime Rule outcomes.
 - Behaviour SHALL be reproducible from the Authority Receipt.
-

Conformance Requirements

A conforming Runtime Authority:

- SHALL verify intended use immediately before execution.
 - SHALL evaluate only Runtime Context defined by this specification.
 - SHALL preserve Runtime Context during evaluation.
 - SHALL record the Runtime Rule outcome within the Authority Receipt.
 - SHALL NOT infer intended use from implementation-specific information.
-

Runtime Rule Summary

Runtime Rule RA-AS-005 confirms that the requested execution remains within the authorised intended use of the AI-enabled ambient scribing system. By preventing execution outside the approved operational scope, the Runtime Authority ensures that delegated authority is exercised only within the boundaries for which it remains valid.

7.8 RA-AS-006 — Operational Context Valid

Identifier

RA-AS-006

Name

Operational Context Valid

Purpose

The purpose of Runtime Rule **RA-AS-006** is to determine whether the operational environment remains suitable for execution immediately before institutional consequence formation.

Unlike workflow or approval validation, this Runtime Rule evaluates the current operational conditions that may affect the legitimacy of execution. It ensures that execution is not performed when observable operational conditions indicate that delegated authority should no longer be exercised.

The Runtime Authority SHALL evaluate only the Runtime Context presented for the current execution request.

Runtime Context

Runtime Context	Identifier
Operational Context	RC-009

Preconditions

Before Runtime Rule evaluation commences:

- Runtime Context SHALL have been validated.
- Operational Context SHALL be present.
- Runtime Authority evaluation SHALL have commenced.
- A valid Execution Request SHALL exist.

If these conditions are not satisfied, Runtime Rule evaluation SHALL NOT proceed.

Required Inputs

Runtime Context	Data Type	Required
Operational Context	Enumeration	Yes
Operational Status	Enumeration	Yes

Evaluation Logic

The Runtime Authority SHALL evaluate operational conditions according to the following deterministic logic.

IF OperationalContext = VALID

THEN

Rule Outcome = ALLOW

ELSE

Rule Outcome = ESCALATE

This Runtime Rule SHALL NOT return REFUSE.

Where operational conditions cannot currently support execution, execution SHALL remain suspended pending a subsequent Runtime Authority evaluation.

Permitted Outcomes

Outcome	Behaviour
---------	-----------

ALLOW	Operational conditions support execution.
--------------	---

ESCALATE	Operational conditions currently prevent execution.
-----------------	---

Evidence Generated

Successful evaluation SHALL generate:

- Runtime Rule Identifier

- Runtime Rule Version
 - Operational Context
 - Operational Status
 - Evaluation Timestamp
 - Runtime Rule Outcome
-

Authority Receipt Mapping

Authority Receipt Field	Source
-------------------------	--------

Runtime Rule Identifier	RA-AS-006
-------------------------	-----------

Operational Context	RC-009
---------------------	--------

Runtime Rule Outcome	Evaluation Result
----------------------	-------------------

Evaluation Timestamp	Runtime Evaluation Engine
----------------------	---------------------------

Deterministic Behaviour

Runtime Rule RA-AS-006 SHALL satisfy the following behavioural properties.

- Identical Runtime Context SHALL produce identical Runtime Rule outcomes.
 - Runtime Context SHALL remain immutable during evaluation.
 - Evaluation SHALL be deterministic.
 - Behaviour SHALL be implementation independent.
 - Evaluation SHALL be reproducible using the Authority Receipt.
-

Conformance Requirements

A conforming Runtime Authority:

- SHALL evaluate operational conditions immediately before execution.

- SHALL preserve Runtime Context during evaluation.
 - SHALL record the Runtime Rule outcome within the Authority Receipt.
 - SHALL NOT infer operational state from implementation-specific behaviour outside the Runtime Context.
 - SHALL preserve deterministic evaluation behaviour.
-

Runtime Rule Summary

Runtime Rule RA-AS-006 confirms that the observable operational environment continues to support execution immediately before institutional consequence formation. Where operational conditions no longer support execution, the Runtime Authority suspends execution by returning an ESCALATE outcome.

7.9 RA-AS-007 — Workflow Context Valid

Identifier

RA-AS-007

Name

Workflow Context Valid

Purpose

The purpose of Runtime Rule **RA-AS-007** is to determine whether the workflow associated with the execution request remains in a valid state immediately before institutional consequence formation.

The Runtime Authority SHALL verify that the execution request originates from a workflow that remains active, authenticated and operationally consistent at the Execution Bind Point.

This Runtime Rule validates workflow continuity rather than workflow design.

Runtime Context

Runtime Context	Identifier
Consultation Active	RC-001
Clinician Authenticated	RC-002
Workflow Context	RC-007

Preconditions

Before Runtime Rule evaluation commences:

- Runtime Context SHALL have been validated.
- Workflow Context SHALL be available.
- Runtime Authority evaluation SHALL have commenced.
- Consultation SHALL remain active.

If these conditions are not satisfied, Runtime Rule evaluation SHALL NOT proceed.

Required Inputs

Runtime Context	Data Type	Required
Consultation Active	Boolean	Yes
Clinician Authenticated	Boolean	Yes
Workflow Context	Enumeration	Yes

Evaluation Logic

The Runtime Authority SHALL evaluate workflow validity according to the following deterministic logic.

IF ConsultationActive = TRUE

AND

ClinicianAuthenticated = TRUE

AND

WorkflowContext = VALID

THEN

Rule Outcome = ALLOW

ELSE

Rule Outcome = REFUSE

Runtime Rule RA-AS-007 SHALL NOT return ESCALATE.

Where workflow continuity cannot be confirmed, institutional consequence SHALL NOT occur.

Permitted Outcomes

Outcome	Behaviour
---------	-----------

ALLOW	Workflow remains valid.
--------------	-------------------------

REFUSE	Workflow integrity cannot be confirmed.
---------------	---

Evidence Generated

Successful evaluation SHALL generate:

- Runtime Rule Identifier
- Runtime Rule Version
- Consultation Status
- Authentication Status
- Workflow Status
- Runtime Rule Outcome
- Evaluation Timestamp

Authority Receipt Mapping

Authority Receipt Field	Source
Runtime Rule Identifier	RA-AS-007
Consultation Status	RC-001
Authentication Status	RC-002
Workflow Status	RC-007
Runtime Rule Outcome	Evaluation Result
Evaluation Timestamp	Runtime Evaluation Engine

Deterministic Behaviour

Runtime Rule RA-AS-007 SHALL satisfy the following behavioural properties.

- Identical Runtime Context SHALL produce identical Runtime Rule outcomes.
 - Runtime Context SHALL remain immutable during evaluation.
 - Evaluation SHALL be deterministic.
 - Behaviour SHALL be implementation independent.
 - Evaluation SHALL be reproducible from the Authority Receipt.
-

Conformance Requirements

A conforming Runtime Authority:

- SHALL evaluate workflow validity before execution.
 - SHALL evaluate only Runtime Context defined by this specification.
 - SHALL preserve Runtime Context during evaluation.
 - SHALL record Runtime Rule outcomes within the Authority Receipt.
 - SHALL preserve deterministic Runtime Rule behaviour.
-

Runtime Rule Summary

Runtime Rule RA-AS-007 confirms that the execution request originates from a workflow that remains operationally valid immediately before institutional consequence formation. By validating workflow continuity, consultation status and clinician authentication, the Runtime Authority ensures that execution proceeds only where the complete operational workflow remains consistent with the delegated authority being exercised.

7.10 Runtime Rule Integrity

Every Runtime Rule defined by this specification SHALL satisfy the following integrity requirements.

A Runtime Rule SHALL:

- possess a unique Runtime Rule Identifier;
- evaluate only the Runtime Context defined by this specification;
- produce only the permitted Runtime Rule outcomes;
- preserve deterministic evaluation behaviour;
- generate engineering evidence supporting the Runtime Authority Decision;
- contribute to exactly one Runtime Authority evaluation.

A Runtime Rule SHALL NOT:

- modify Runtime Context during evaluation;
- infer Runtime Context not presented at the Execution Bind Point;
- depend upon implementation-specific behaviour not represented within the Runtime Context;
- return outcomes not defined by this specification.

Compliance with these integrity requirements is mandatory for conformance with ASVH-SPEC-001.

7.11 Engineering Summary

The Runtime Rule Specifications define the complete normative behaviour of the Runtime Rule Catalogue.

Collectively, the seven Runtime Rules evaluate the Runtime Context presented at the Execution Bind Point and determine whether delegated authority remains legitimately exercisable immediately before institutional consequence formation. Each Runtime Rule evaluates a distinct observable aspect of the execution request while preserving deterministic behaviour, implementation independence and evidence generation.

Together, the Runtime Rule Catalogue establishes the behavioural contract upon which Runtime Authority evaluation depends. Every conforming Runtime Authority implementation SHALL evaluate Runtime Rules in accordance with this chapter, preserve Runtime Context throughout evaluation, generate the required Authority Receipt evidence and exhibit the deterministic behaviour defined by this specification.

8. Runtime Evaluation Engine

8.1 Overview

The Runtime Evaluation Engine defines the normative behaviour through which the Runtime Authority evaluates Runtime Context and determines a Runtime Authority Decision.

A conforming Runtime Authority **SHALL** evaluate Runtime Rules using the Runtime Evaluation Engine defined by this specification.

The Runtime Evaluation Engine SHALL operate deterministically and SHALL produce identical Runtime Authority Decisions whenever identical Runtime Context is presented.

8.2 Engineering Responsibility

The Runtime Evaluation Engine **SHALL**:

- receive a validated Runtime Context;
- determine the applicable Runtime Rule set;
- evaluate Runtime Rules according to this specification;
- collect Runtime Rule outcomes;
- determine a single Runtime Authority Decision;
- generate an Authority Receipt.

The Runtime Evaluation Engine SHALL NOT modify Runtime Context during evaluation.

8.3 Runtime Evaluation Lifecycle

Runtime Authority evaluation SHALL occur using the following lifecycle.

1. Runtime Context received.
2. Runtime Context validated.
3. Applicable Runtime Rules identified.
4. Runtime Rules evaluated.
5. Runtime Rule outcomes collected.
6. Runtime Authority Decision determined.
7. Authority Receipt generated.
8. Runtime Authority Decision returned.

Every Runtime Authority evaluation SHALL follow this sequence.

8.4 Rule Evaluation Order

A conforming Runtime Authority SHALL evaluate Runtime Rules in the following normative order.

Order	Runtime Rule
-------	--------------

- 1 RA-AS-001 Target Commit Path Available
- 2 RA-AS-004 Execution Type Identified
- 3 RA-AS-007 Workflow Context Valid
- 4 RA-AS-006 Operational Context Valid
- 5 RA-AS-005 Intended Use Maintained
- 6 RA-AS-002 Output Reviewed
- 7 RA-AS-003 Output Approved

A conforming Runtime Authority SHALL produce behaviour equivalent to the evaluation model defined by this specification.

8.5 Independent Runtime Rule Evaluation

Each Runtime Rule SHALL be evaluated independently.

Evaluation of one Runtime Rule SHALL NOT modify:

- Runtime Context;
- Runtime Rule definitions;
- Runtime Rule outcomes;
- evaluation behaviour of any other Runtime Rule.

This requirement preserves deterministic Runtime Authority behaviour.

8.6 Outcome Aggregation

Following completion of Runtime Rule evaluation, the Runtime Evaluation Engine SHALL aggregate all Runtime Rule outcomes.

Outcome aggregation SHALL determine one Runtime Authority Decision.

The aggregation process SHALL be deterministic.

Identical Runtime Rule outcomes SHALL always produce identical Runtime Authority Decisions.

8.7 Evaluation Termination

Runtime Authority evaluation SHALL terminate when one of the following conditions is satisfied.

Successful Evaluation

All applicable Runtime Rules have completed evaluation.

A Runtime Authority Decision SHALL be returned.

REFUSE Outcome

Where a Runtime Rule produces a REFUSE outcome, evaluation MAY terminate immediately, provided that:

- the Runtime Authority Decision is determined as REFUSE;
 - the Authority Receipt records all Runtime Rule evaluations completed prior to termination;
 - the terminating Runtime Rule is identified.
-

ESCALATE Outcome

An ESCALATE outcome SHALL suspend execution.

The Runtime Authority SHALL return ESCALATE until a subsequent evaluation produces an alternative Runtime Authority Decision.

8.8 Evidence Collection

Throughout Runtime Rule evaluation, the Runtime Evaluation Engine SHALL collect engineering evidence including:

- Runtime Context evaluated;
- Runtime Rule identifiers;
- Runtime Rule outcomes;
- Runtime Authority Decision;
- Evaluation Timestamp;
- Runtime Authority Version.

Collected evidence SHALL be incorporated into the Authority Receipt.

8.9 Deterministic Behaviour

The Runtime Evaluation Engine SHALL satisfy the following behavioural properties.

Property	Requirement
Deterministic	Identical Runtime Context SHALL produce identical Runtime Authority Decisions.
Independent	Runtime Rules SHALL evaluate independently.
Reproducible	Runtime evaluations SHALL be capable of deterministic replay.
Immutable	Runtime Context SHALL remain unchanged during evaluation.
Explainable	Runtime Rule outcomes SHALL be represented within the Authority Receipt.

8.10 Relationship to Companion Publications

This chapter defines the normative behaviour of the Runtime Evaluation Engine.

Supporting engineering information is provided by the companion publications.

Publication	Relationship
ASVH-METH-001	Describes the methodology used to derive Runtime Rules.
ASVH-TRACE-001	Defines the engineering relationships between Runtime Evaluation, Runtime Rules and Authority Receipts.
ASVH-ARCH-001	Describes the architectural placement of the Runtime Evaluation Engine.
ASVH-IMPL-001	Demonstrates one conforming implementation of the Runtime Evaluation Engine.

8.11 Engineering Summary

The Runtime Evaluation Engine defines the normative execution behaviour of the Runtime Authority.

By evaluating Runtime Rules in a deterministic sequence, preserving Runtime Context, aggregating Runtime Rule outcomes and generating an Authority Receipt, the Runtime Evaluation Engine provides the implementation-independent mechanism through which Runtime Authority Decisions are consistently produced.

9. Runtime Authority Decision Model

9.1 Overview

The Runtime Authority Decision Model defines the complete set of Runtime Authority Decisions permitted by this specification.

Following completion of Runtime Rule evaluation, a conforming Runtime Authority **SHALL** determine exactly one Runtime Authority Decision.

No additional Runtime Authority Decision values are permitted.

9.2 Decision Formation

A Runtime Authority Decision SHALL be determined only after completion of Runtime Rule evaluation in accordance with Chapter 8.

The Runtime Authority SHALL consider the complete set of Runtime Rule outcomes before determining the final Runtime Authority Decision.

A Runtime Authority Decision SHALL:

- be deterministic;
 - be derived exclusively from Runtime Rule outcomes;
 - represent the outcome of the current execution request only;
 - be recorded within the Authority Receipt.
-

9.3 Permitted Runtime Authority Decisions

This specification recognises exactly three Runtime Authority Decisions.

Decision	Definition	Execution Behaviour
ALLOW	Delegated authority remains legitimately exercisable.	Execution MAY proceed.
ESCALATE	Runtime evaluation cannot currently authorise execution.	Execution SHALL remain suspended pending further action.
REFUSE	Delegated authority is no longer legitimately exercisable.	Execution SHALL NOT proceed.

No additional Runtime Authority Decision values SHALL be used by a conforming implementation.

9.4 ALLOW

An **ALLOW** decision indicates that the Runtime Authority has determined that delegated authority remains legitimately exercisable for the presented Runtime Context.

A conforming Runtime Authority **SHALL** return **ALLOW** only where the applicable Runtime Rule outcomes collectively support continued execution.

An **ALLOW** decision:

- permits downstream execution;
 - SHALL be recorded within the Authority Receipt;
 - SHALL identify the Runtime Rule outcomes supporting the decision.
-

9.5 ESCALATE

An **ESCALATE** decision indicates that the Runtime Authority cannot currently determine that execution is admissible.

Execution SHALL remain suspended until a subsequent Runtime Authority evaluation determines an alternative decision.

An **ESCALATE** decision:

- SHALL prevent downstream execution;
- SHALL preserve the Runtime Context evaluated;
- SHALL generate an Authority Receipt;
- SHALL identify the Runtime Rule outcome(s) requiring escalation.

This specification does not prescribe the operational activities that occur following escalation.

9.6 REFUSE

A **REFUSE** decision indicates that delegated authority is no longer legitimately exercisable for the presented Runtime Context.

A conforming Runtime Authority **SHALL** prevent institutional consequence formation following a **REFUSE** decision.

A **REFUSE** decision:

- SHALL terminate the current execution request;
- SHALL generate an Authority Receipt;
- SHALL preserve the Runtime Rule outcomes contributing to the decision.

Subsequent execution requires a new Runtime Authority evaluation.

9.7 Decision Integrity

Every Runtime Authority Decision SHALL satisfy the following requirements.

Requirement	Behaviour
Unique	Exactly one decision SHALL be returned.
Deterministic	Identical Runtime Rule outcomes SHALL produce identical decisions.
Traceable	Every decision SHALL be represented within the Authority Receipt.
Immutable	The recorded decision SHALL NOT be modified following completion of evaluation.
Reproducible	The decision SHALL be capable of deterministic reconstruction using the recorded Runtime Context and Runtime Rule outcomes.

9.8 Prohibited Behaviour

A conforming Runtime Authority **SHALL NOT**:

- return multiple Runtime Authority Decisions for a single evaluation;
- return undefined decision values;
- modify a Runtime Authority Decision after Authority Receipt generation;
- infer Runtime Authority Decisions from information outside the Runtime Context;
- bypass Runtime Rule evaluation when determining a decision.

9.9 Relationship to Companion Publications

This chapter defines the normative Runtime Authority Decision Model.

Supporting engineering information is provided by the companion publications.

Publication	Relationship
ASVH-METH-001	Describes the engineering methodology used to derive Runtime Rules.
ASVH-TRACE-001	Defines the engineering relationships linking Runtime Authority Decisions to Runtime Rules and Authority Receipts.
ASVH-ARCH-001	Describes the architectural location of Runtime Authority decision formation.
ASVH-IMPL-001	Demonstrates one conforming Runtime Authority implementation.

9.10 Engineering Summary

The Runtime Authority Decision Model defines the complete set of deterministic outcomes recognised by this specification.

Every conforming Runtime Authority SHALL determine exactly one Runtime Authority Decision following Runtime Rule evaluation. The resulting decision SHALL be deterministic, implementation independent and fully represented within the Authority Receipt, providing the authoritative basis for subsequent execution behaviour.

10. Authority Receipt Specification

10.1 Overview

The Authority Receipt is the normative engineering record produced by the Runtime Authority following completion of every Runtime Authority evaluation.

A conforming Runtime Authority **SHALL** generate exactly one Authority Receipt for every completed Runtime Authority evaluation, irrespective of the resulting Runtime Authority Decision.

The Authority Receipt provides the implementation-independent engineering evidence required to reconstruct the completed Runtime Authority evaluation.

10.2 Engineering Purpose

The Authority Receipt SHALL:

- preserve the Runtime Context presented for evaluation;
- record the Runtime Rules evaluated;
- record the Runtime Rule outcomes;
- record the Runtime Authority Decision;
- preserve engineering evidence supporting deterministic reconstruction.

The Authority Receipt SHALL represent the authoritative record of Runtime Authority evaluation.

10.3 Authority Receipt Characteristics

Every Authority Receipt SHALL exhibit the following characteristics.

Characteristic	Requirement
Complete	SHALL contain all mandatory receipt fields defined by this specification.

Deterministic	SHALL accurately represent the completed Runtime Authority evaluation.
Immutable	SHALL NOT be modified after generation.
Traceable	SHALL identify the Runtime Context and Runtime Rules evaluated.
Specification Version	Version of ASVH-SPEC-001 evaluated against
Implementation Independent	SHALL preserve identical engineering meaning regardless of implementation technology.

10.4 Mandatory Authority Receipt Fields

Every Authority Receipt SHALL contain the following mandatory fields.

Field	Type	Required	Description
Receipt Identifier	UUID	Yes	Unique Authority Receipt identifier.
Evaluation Timestamp	DateTime	Yes	Time at which Runtime Authority evaluation completed.
Runtime Authority Version	String	Yes	Version of the Runtime Authority implementation.

Runtime Authority Decision	Enumeration	Yes	ALLOW, ESCALATE or REFUSE.
Runtime Context Identifier	Collection	Yes	Runtime Context evaluated.
Runtime Rule Identifiers	Collection	Yes	Runtime Rules evaluated.
Runtime Rule Outcomes	Collection	Yes	Outcome of each Runtime Rule evaluation.
Evidence Integrity Value	String	Yes	Integrity value supporting evidence verification.

These fields constitute the minimum Authority Receipt required for conformance.

10.5 Optional Authority Receipt Fields

A conforming Runtime Authority **MAY** include additional implementation-specific fields.

Examples include:

- transaction identifier;
- workflow identifier;
- system identifier;
- implementation metadata;
- diagnostic information.

Optional fields SHALL NOT alter the engineering meaning of the mandatory Authority Receipt fields defined by this specification.

10.6 Receipt Generation

Authority Receipt generation SHALL occur immediately following determination of the Runtime Authority Decision.

A conforming Runtime Authority SHALL:

1. complete Runtime Rule evaluation;
2. determine the Runtime Authority Decision;
3. assemble mandatory Authority Receipt fields;
4. preserve Runtime Rule outcomes;
5. preserve Runtime Context references;
6. generate the Authority Receipt;
7. return the Runtime Authority Decision.

Authority Receipt generation SHALL occur exactly once for every completed Runtime Authority evaluation.

10.7 Evidence Integrity

The Authority Receipt SHALL preserve sufficient engineering evidence to enable deterministic reconstruction of the completed Runtime Authority evaluation.

A conforming Runtime Authority SHALL preserve:

- Runtime Context references;
- Runtime Rule identifiers;
- Runtime Rule outcomes;
- Runtime Authority Decision;
- evaluation timestamp;
- Authority Receipt identifier.

A conforming implementation **MAY** additionally preserve integrity values, digital signatures or cryptographic attestations. Such mechanisms are implementation-specific enhancements and are **not** required for conformance with this specification.

10.8 Authority Receipt Lifecycle

The Authority Receipt lifecycle consists of five stages.

1. Evaluation completed.
2. Runtime Authority Decision determined.
3. Authority Receipt assembled.
4. Authority Receipt finalised.
5. Authority Receipt preserved.

Once finalised, the Authority Receipt SHALL remain immutable.

10.9 Prohibited Behaviour

A conforming Runtime Authority **SHALL NOT**:

- generate multiple Authority Receipts for a single Runtime Authority evaluation;
 - modify mandatory Authority Receipt fields after generation;
 - omit mandatory Authority Receipt fields;
 - record Runtime Context that was not evaluated;
 - alter Runtime Rule outcomes following receipt generation.
-

10.10 Relationship to Companion Publications

This chapter defines the normative Authority Receipt specification.

Supporting engineering information is provided by the companion publications.

Publication	Relationship
ASVH-METH-001	Defines the engineering methodology used to derive Runtime Context and Runtime Rules.
ASVH-TRACE-001	Defines the engineering relationships linking Authority Receipts to Runtime Context, Runtime Rules and Verification Scenarios.
ASVH-ARCH-001	Describes the architectural location of Authority Receipt generation.
ASVH-IMPL-001	Demonstrates one conforming Authority Receipt implementation.

10.11 Engineering Summary

The Authority Receipt Specification defines the minimum engineering evidence that every conforming Runtime Authority **SHALL** generate following Runtime Authority evaluation.

By specifying mandatory receipt content, evidence integrity requirements and receipt lifecycle behaviour independently of implementation technology, this chapter establishes the Authority Receipt as the authoritative and immutable engineering record supporting deterministic Runtime Authority evaluation.

11. Conformance Requirements

11.1 Overview

This chapter defines the minimum behavioural requirements that a Runtime Authority implementation SHALL satisfy to claim conformance with ASVH-SPEC-001.

Conformance is determined by observable Runtime Authority behaviour rather than software architecture, implementation technology or deployment model.

A Runtime Authority implementation SHALL demonstrate that it preserves the deterministic behaviour defined throughout this specification.

11.2 Conformance Principles

A conforming Runtime Authority implementation SHALL satisfy the following principles:

- deterministic Runtime Rule evaluation;
- implementation-independent behaviour;
- immutable Runtime Context evaluation;
- deterministic Runtime Authority Decision formation;
- Authority Receipt generation;
- evidence preservation.

Failure to satisfy any mandatory requirement defined by this specification SHALL constitute non-conformance.

11.3 Mandatory Requirements

A conforming Runtime Authority implementation **SHALL**:

1. accept a valid Runtime Context as defined in Chapter 5;
2. evaluate Runtime Rules defined in Chapter 7;

3. evaluate Runtime Rules independently;
4. preserve Runtime Context during evaluation;
5. produce exactly one Runtime Authority Decision;
6. generate exactly one Authority Receipt;
7. preserve Authority Receipt integrity;
8. produce deterministic Runtime Authority behaviour;
9. preserve implementation-independent engineering semantics.

These requirements constitute the minimum mandatory behaviour required for conformance.

11.4 Recommended Behaviour

A conforming Runtime Authority implementation **SHOULD**:

- support deterministic replay of Runtime Authority evaluations;
- preserve complete Authority Receipt history;
- expose Runtime Rule outcomes for engineering review;
- preserve Runtime Authority version information;
- support engineering traceability to Runtime Context identifiers.

These behaviours improve engineering assurance but are not mandatory for conformance.

11.5 Optional Behaviour

A conforming Runtime Authority implementation **MAY**:

- implement additional Runtime Rules beyond those defined in this specification;
- extend the Authority Receipt with implementation-specific metadata;
- support alternative persistence mechanisms;
- support multiple deployment architectures;
- integrate with external evidence repositories.

Optional behaviour **SHALL NOT** modify the engineering meaning of the Runtime Authority behaviour defined by this specification.

11.6 Prohibited Behaviour

A conforming Runtime Authority implementation **SHALL NOT**:

- modify Runtime Context during evaluation;
- bypass Runtime Rule evaluation;
- infer Runtime Authority Decisions using information outside the Runtime Context;
- generate multiple Runtime Authority Decisions for a single evaluation;
- modify Authority Receipts following generation;
- omit mandatory Runtime Rule evaluations;
- return Runtime Authority Decisions not defined by this specification.

Implementations exhibiting any prohibited behaviour SHALL NOT claim conformance.

11.7 Conformance Assessment

Conformance SHALL be assessed by observing Runtime Authority behaviour.

Assessment SHALL confirm that:

- Runtime Context is evaluated according to this specification;
- Runtime Rules behave deterministically;
- Runtime Authority Decisions conform to Chapter 9;
- Authority Receipts satisfy Chapter 10;
- Prohibited behaviour is absent.

Acceptable evidence sources

- Deterministic replay;
- Verification scenarios
- Authority Receipt inspection
- Implementation testing
- Engineering review

Conformance assessment is independent of software architecture, programming language or deployment technology.

11.8 Relationship to Companion Publications

This chapter defines the conformance requirements for Runtime Authority implementations.

Supporting engineering information is provided by the companion publications.

Publication	Relationship
ASVH-METH-001	Defines the engineering methodology used to derive Runtime Requirements and Runtime Rules.
ASVH-TRACE-001	Defines the engineering relationships supporting conformance assessment.
ASVH-ARCH-001	Describes one architectural implementation preserving conformance.
ASVH-IMPL-001	Demonstrates a conforming Runtime Authority implementation.

11.9 Engineering Summary

This specification defines behavioural conformance rather than implementation conformance.

A Runtime Authority implementation conforms to ASVH-SPEC-001 by demonstrating deterministic Runtime Authority behaviour, preserving Runtime Context integrity, evaluating the Runtime Rule Catalogue, generating Authority Receipts and satisfying the mandatory requirements defined throughout this specification.

Conformance is therefore determined by engineering behaviour rather than implementation technology.

12. Conclusion

12.1 Specification Summary

ASVH-SPEC-001 defines the normative behaviour required of a Runtime Authority operating at the Execution Bind Point immediately before institutional consequence formation.

The specification establishes the contractual engineering behaviour that every conforming Runtime Authority implementation SHALL exhibit, independently of software architecture, implementation technology or deployment model.

By defining Runtime Context, Runtime Rules, Runtime Evaluation, Runtime Authority Decisions, Authority Receipts and Conformance Requirements, this specification provides a complete implementation-independent behavioural contract for deterministic Runtime Authority evaluation.

12.2 Engineering Contribution

The principal contribution of this specification is the definition of a deterministic Runtime Authority behavioural model.

Rather than prescribing implementation technology, the specification defines the engineering semantics that SHALL be preserved by every conforming implementation.

The Runtime Authority defined by this specification:

- evaluates Runtime Context presented at the Execution Bind Point;
- applies the Runtime Rule Catalogue deterministically;
- determines a single Runtime Authority Decision;

- generates an Authority Receipt recording the completed evaluation.

These behaviours collectively establish a repeatable engineering mechanism for determining whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

12.3 Relationship to the ASVH Publication Series

ASVH-SPEC-001 forms the normative behavioural specification within the ASVH engineering publication series.

Each publication addresses a distinct engineering responsibility.

Publication	Engineering Responsibility
ASVH-METH-001	Defines the engineering methodology used to derive Runtime Requirements and Runtime Rules.
ASVH-TRACE-001	Defines the engineering relationships linking Runtime Verification artefacts.
ASVH-SPEC-001	Defines the normative behavioural contract for Runtime Authority implementations.
ASVH-ARCH-001	Defines the architectural placement of the Runtime Authority.
ASVH-IMPL-001	Demonstrates one conforming Runtime Authority implementation.

Collectively, these publications provide a complete engineering framework for deterministic Runtime Verification while maintaining clear separation between methodology, specification, traceability, architecture and implementation.

12.4 Implementation Independence

This specification intentionally defines behaviour rather than implementation.

Conformance is determined by the observable behaviour of the Runtime Authority and not by the software architecture, programming language, infrastructure or deployment model used to implement it.

Accordingly, multiple independent implementations may conform to this specification provided they preserve the deterministic Runtime Authority behaviour defined throughout this document.

12.5 Final Statement

This specification establishes the Runtime Authority as an implementation-independent engineering component responsible for determining whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

By defining deterministic Runtime Rule behaviour, Runtime Authority Decision formation, Authority Receipt generation and behavioural conformance requirements, ASVH-SPEC-001 provides the normative engineering contract against which Runtime Authority implementations can be objectively assessed.

Together with the Runtime Requirement Engineering methodology, the Engineering Traceability Model, the Reference Architecture and the Reference Implementation, this specification completes the engineering foundation of the ASVH Runtime Verification Framework and provides a consistent, implementation-independent basis for deterministic Runtime Authority evaluation.

